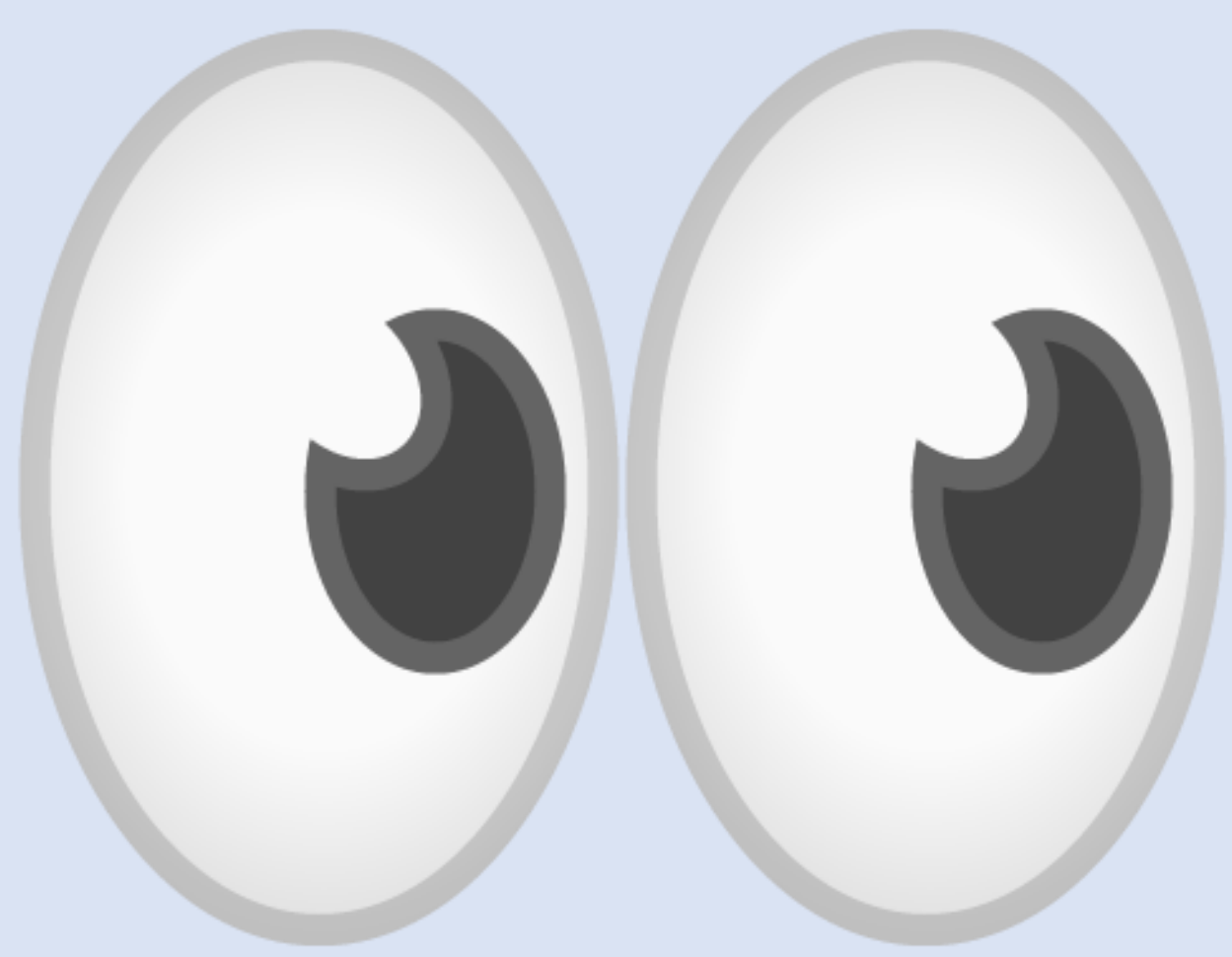




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EyeBench: Predictive Modeling from Eye Movements in Reading

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University of Zurich



University of Potsdam

A standardized framework for decoding cognitive and linguistic information from eye movements during reading

Tasks & Datasets

Data Harmonization

Modeling

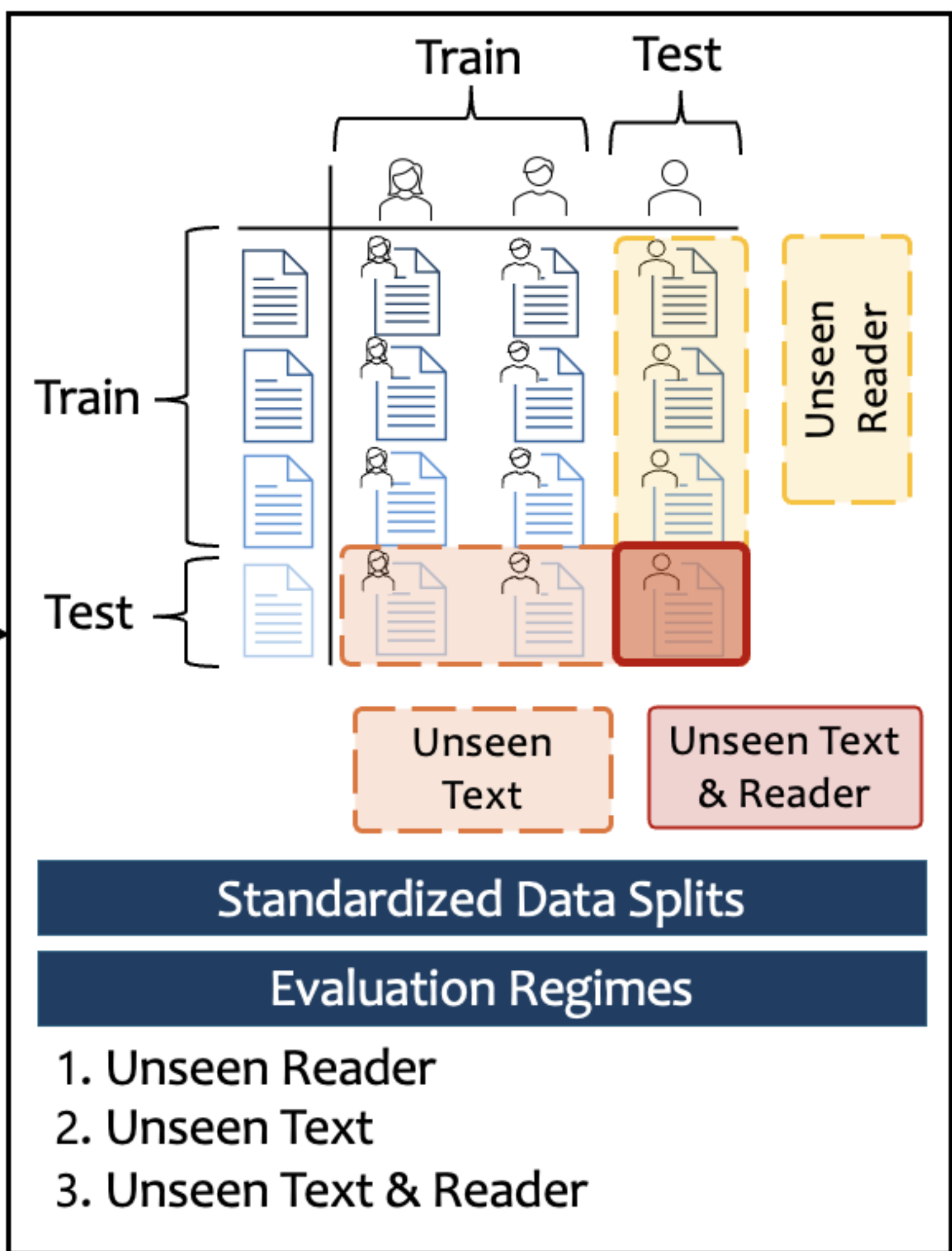
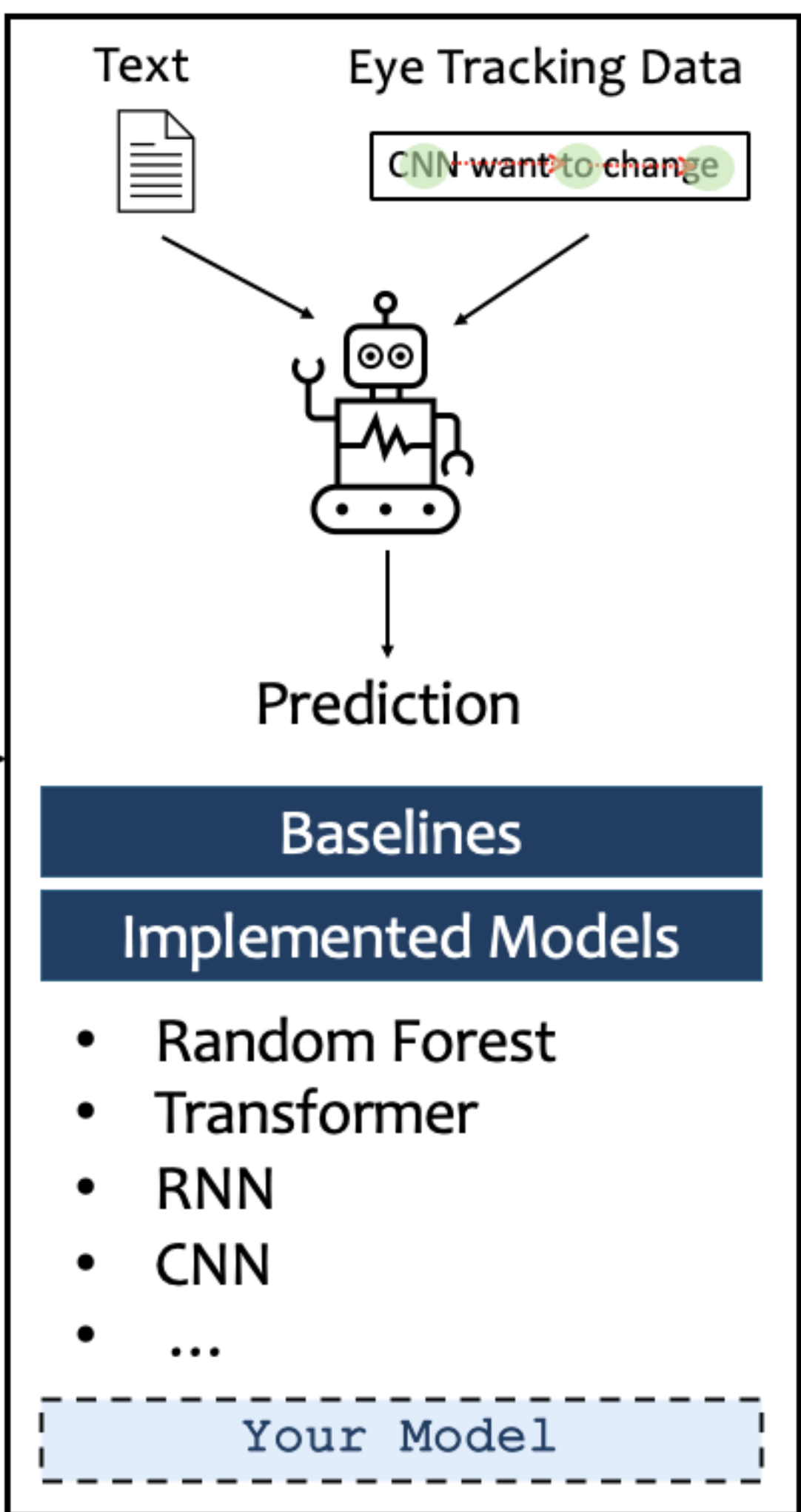
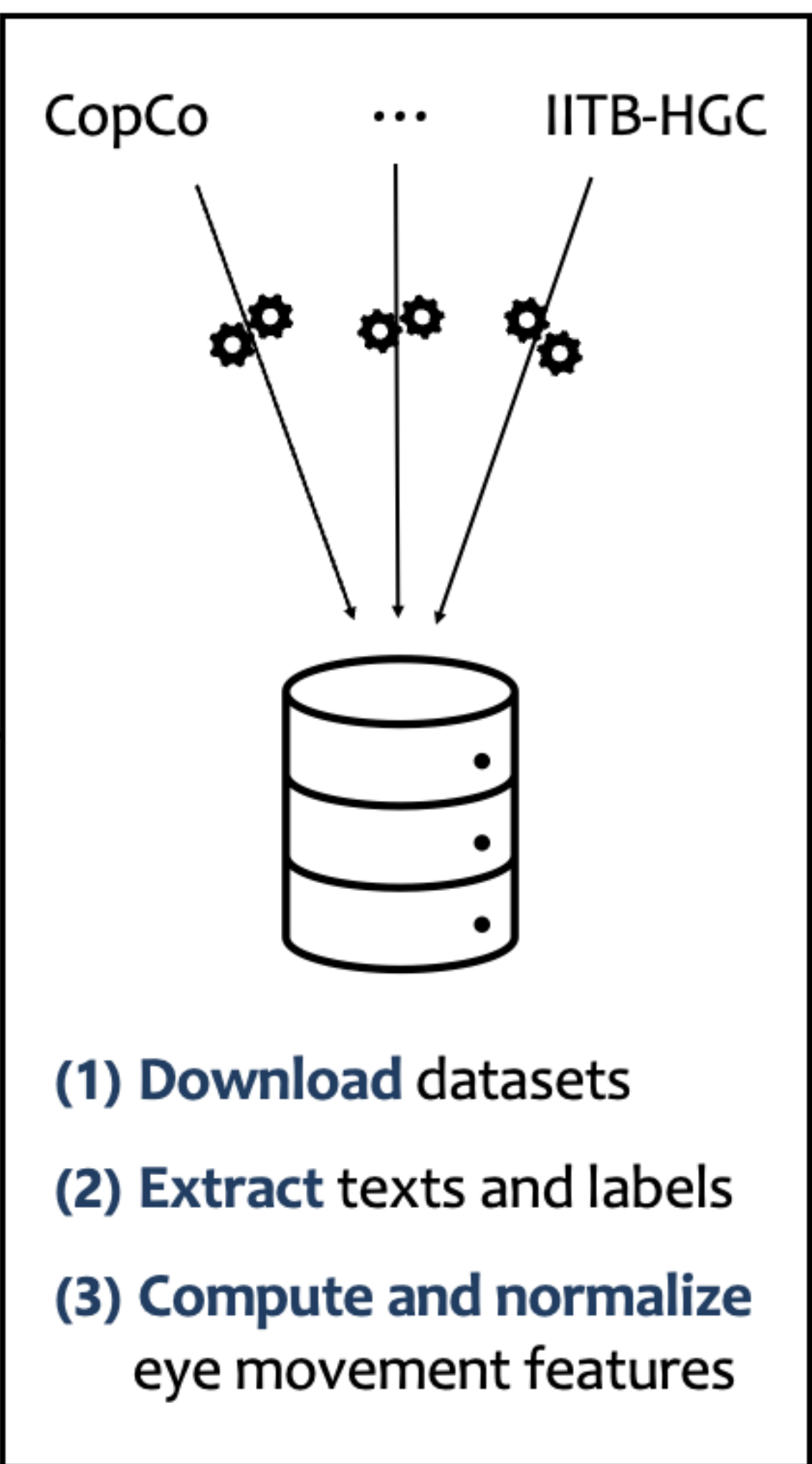
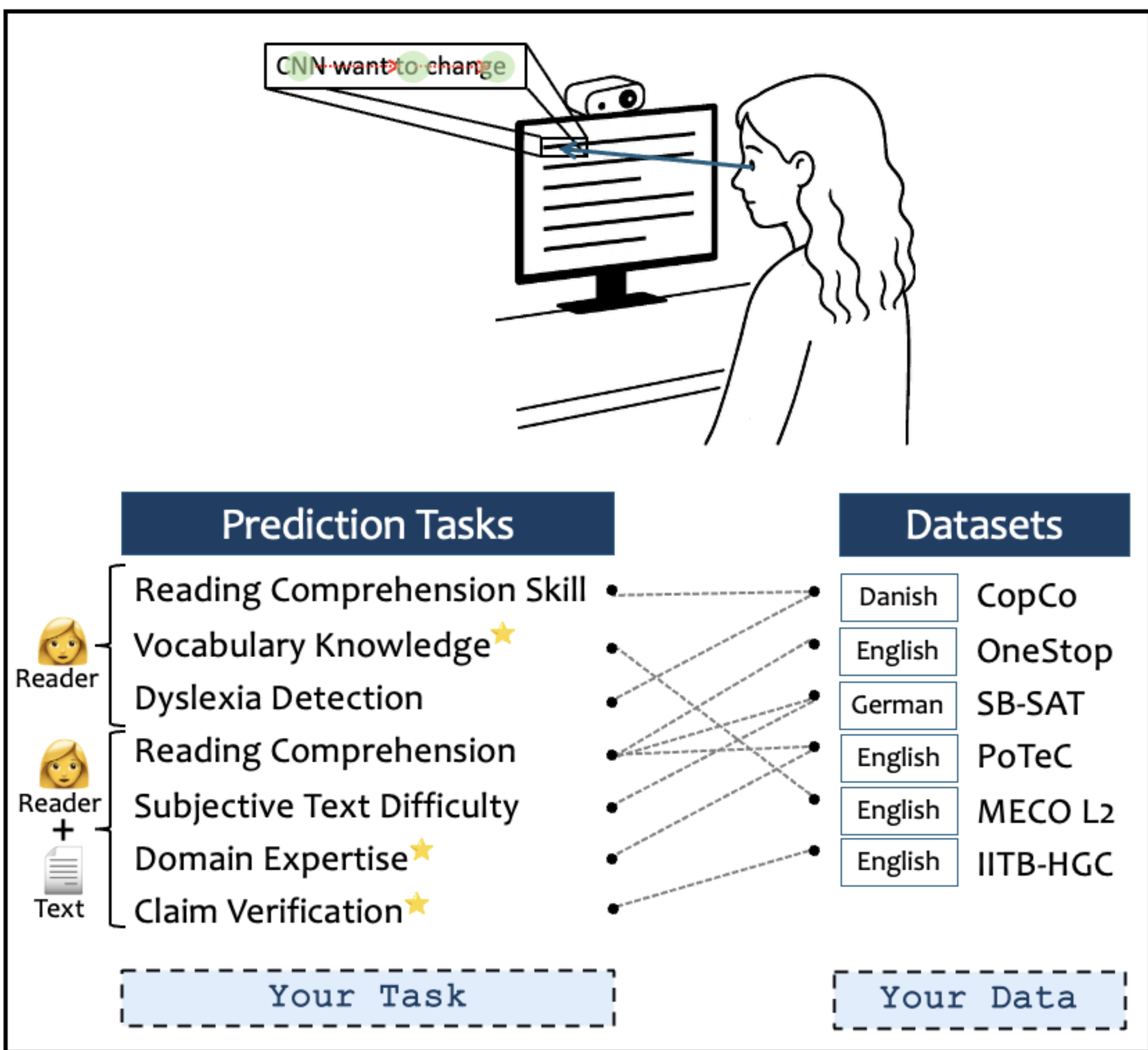
Evaluation

7 Challenging Tasks based on 6 Eye Tracking Datasets

Preprocessed and easy to use

15 Models

3 Evaluation Regimes



~1.5K
Participants

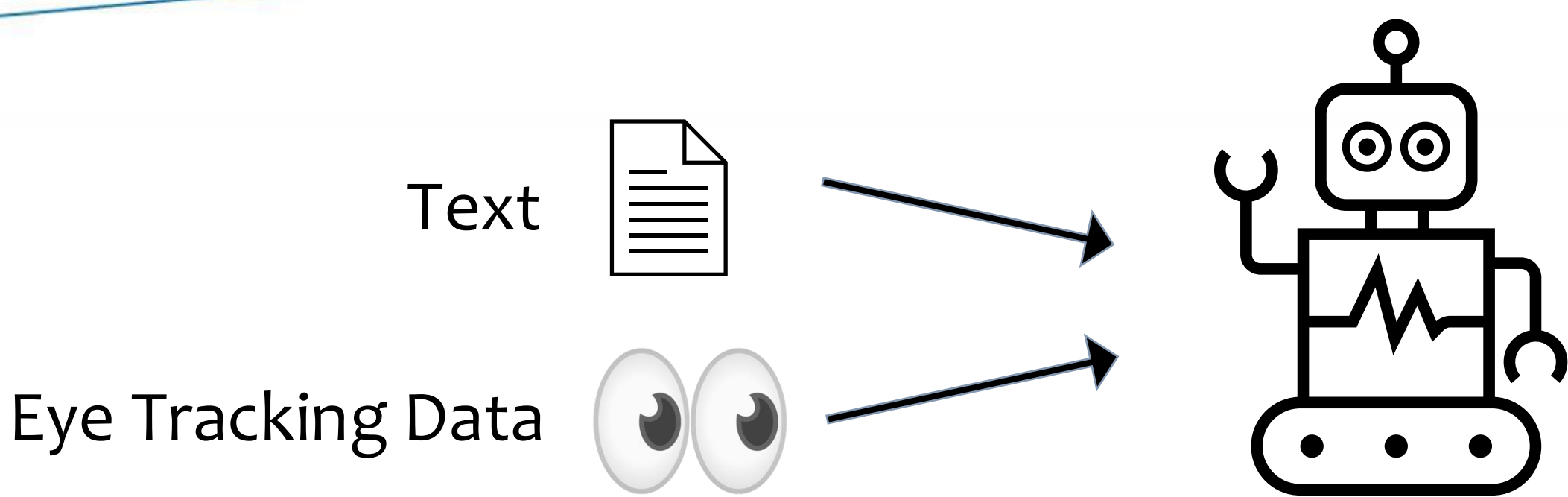
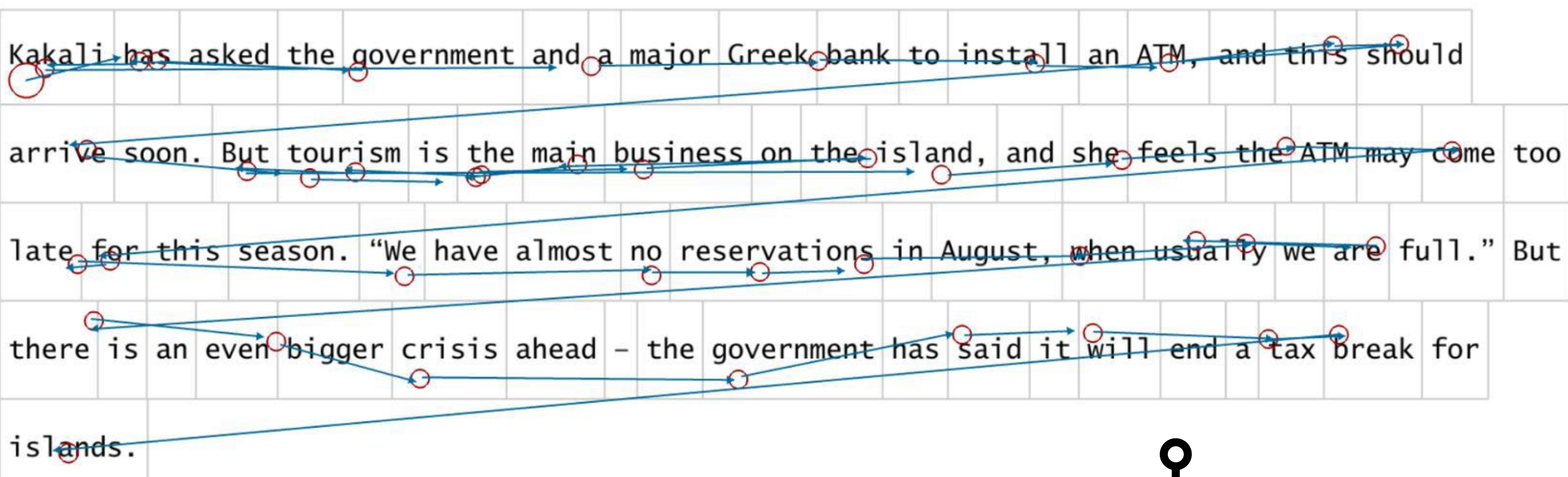
~4.7M
Fixations

~110K
Words

~31K
Reading Trials

Underexplored Modelling Challenge

EyeBench V1.0 Results



Model	Layout	Eye movement features			Text features		Aggregated performance	
		Saccade/ Fixation Level	Word Level	Trial Level	Linguistic	Embeddings	Avg. Normalized Score↑	Mean Rank↓
Majority Class / Chance	-	-	-	-	-	-	0.211	12.38
Reading Speed	-	-	-	✓	-	-	0.376	10.05
Text-Only Roberta	-	-	-	-	-	✓	0.641	6.05
Logistic Regression [31]	-	-	-	✓	-	-	0.491	8.90
SVM [73]	-	-	-	✓	-	-	0.502	8.76
XGBoost	-	-	-	✓	✓	-	0.719	4.86
Random Forest [74]	-	✓	-	✓	✓	-	0.730	4.52
AhnRNN [37]	✓	✓	-	-	-	-	0.217	12.19
AhnCNN [37]	✓	✓	-	-	-	-	0.500	8.33
BEyeLSTM [27]	✓	✓	✓	✓	✓	-	0.348	10.00
PLM-AS [72]	-	✓	✓	-	-	✓	0.467	8.57
PLM-AS-RM [57]	-	✓	✓	-	-	✓	0.605	6.90
RoBERTa-Eye-W [32]	✓	-	✓	✓	✓	✓	0.658	5.43
RoBERTa-Eye-F [32]	✓	-	✓	✓	✓	✓	0.494	8.57
MAG-Eye [32]	✓	-	✓	✓	✓	✓	0.580	6.62
PostFusion-Eye [32]	✓	✓	✓	✓	✓	✓	0.517	8.76

Get Involved!

Full results, contribution guidelines and more at

[EyeBench.github.io](https://github.com/eye-bench/eye-bench)